In [ ]:
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 Experiment 5: Data Visualization with Matplotlib

PIN: 241UDS0097

Aim: To visualize and analyze a practice dataset using Matplotlib plots and
identify trends in LPG production, consumption, imports, prices, and
subsidies.
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## Step 1: Import the required libraries and dataset
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In [ ]:
```
import pandas as pd
import matplotlib.pyplot as plt

df_241UDS0097 = pd.read_csv('Experiment_5_241UDS0097_India_LPG_Dataset.csv')
df_241UDS0097.info()
```

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## Step 2: Display the dataset
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In [ ]:
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df_241UDS0097.head()
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df_241UDS0097
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In [ ]:
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df_241UDS0097.columns
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In [ ]:
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# Convert Year into a suitable numeric format
df_241UDS0097['Year'] = pd.to_numeric(df_241UDS0097['Year'])
df_241UDS0097['Year'].dtype
```

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## Step 3: Visualize data using various plots
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In [ ]:
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### 1. LPG Price Over Time
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In [ ]:
```
plt.figure(figsize=(15, 6), dpi=150)
plt.plot(df_241UDS0097['Year'], df_241UDS0097['Price_INR'], marker='x',
label='LPG Price')
plt.title('LPG Price Over Time')
plt.xlabel('Year')
plt.ylabel('Price (INR)')
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```

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### 2. Subplots: LPG Consumption and Production
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In [ ]:
```
fig, axes = plt.subplots(1, 2, figsize=(16, 4))

axes[0].scatter(df_241UDS0097['Year'], df_241UDS0097['Consumption_MMT'])
axes[0].set_title('LPG Consumption Over Time')
axes[0].set_xlabel('Year')
axes[0].set_ylabel('Consumption (MMT)')

axes[1].scatter(df_241UDS0097['Year'], df_241UDS0097['Production_MMT'])
axes[1].set_title('LPG Production Over Time')
axes[1].set_xlabel('Year')
axes[1].set_ylabel('Production (MMT)')

plt.tight_layout()
plt.show()
```

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### 3. Comparing LPG Production, Consumption and Imports
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In [ ]:
```
plt.figure(figsize=(15, 6), dpi=150)
plt.plot(df_241UDS0097['Year'], df_241UDS0097['Production_MMT'],
label='Production')
plt.plot(df_241UDS0097['Year'], df_241UDS0097['Consumption_MMT'],
label='Consumption')
plt.plot(df_241UDS0097['Year'], df_241UDS0097['Imports_MMT'],
label='Imports')
plt.title('LPG Production, Consumption and Imports Over Time')
plt.xlabel('Year')
plt.ylabel('Quantity (MMT)')
plt.tight_layout()
plt.legend(loc='upper left', title='LPG Parameters')
plt.grid(True, linestyle='--')
plt.show()
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### 4. LPG Price vs Crude Oil Price
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In [ ]:
```
plt.figure(figsize=(10, 6), dpi=150)
plt.scatter(df_241UDS0097['Crude_Price_USD'], df_241UDS0097['Price_INR'])
plt.title('LPG Price vs Crude Oil Price')
plt.xlabel('Crude Oil Price (USD)')
plt.ylabel('LPG Price (INR)')
plt.grid(True)
plt.show()
```

In [ ]:
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### 5. LPG Subsidy Over Time
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In [ ]:
```
plt.figure(figsize=(15, 6), dpi=150)
plt.plot(df_241UDS0097['Year'], df_241UDS0097['Subsidy_Crore'], marker='o',
label='Subsidy')
plt.title('LPG Subsidy Over Time')
plt.xlabel('Year')
plt.ylabel('Subsidy (Crore)')
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```

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Result:-

Various data visualization techniques using Matplotlib are implemented to
inspect practice-data trends in LPG production, consumption, imports,
prices, and subsidy over time.
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